Does Cost-Efficiency Travel? Rank Transfer and Frontier Stability in Fixed-Scaffold Agent Evaluation

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Abstract

Agent leaderboards increasingly report task success and dollar cost, inviting practitioners to select a "cost-efficient" system from a single benchmark. We test whether this signal transfers across workloads and whether any transfer reflects stable model-label cost propensity. Using public Holistic Agent Leaderboard (HAL) results, we hold the displayed HAL Generalist Agent scaffold fixed and compare repeated displayed model labels across six benchmarks. Raw dollar-cost ranks transfer robustly in several high-overlap pairs: among pairs with at least 12 shared labels, mean pairwise Spearman correlation is 0.84. A pooled log-cost model with benchmark and displayed-label fixed effects, fit with 30 parameters on 86 rows (56 residual degrees of freedom), explains 82 percent of observed log-cost variance; this descriptive in-sample quantity is not our primary evidence. In LOBO prediction, displayed-label cost propensity learned from other workloads predicts raw cost ranks on four held-out workloads and fails on two lower-overlap workloads. Aggregate CostPerSuccess, dollar cost divided by max(accuracy, 1 percent), shows three Holm-supported positive pairwise associations, while most other CPS associations remain uncertain. Finally, an external matched unencrypted trace sample of 977 same-task/same-model pairs shows that OpenHands uses fewer median turns and tool calls than SWE-agent without a detected resolved-status difference. Thus dollar cost mixes stable label propensity and scaffold trajectory structure; raw cost transfer alone is not portable deployment efficiency. No broad label lies on every cost–accuracy frontier under three definitions.

1 Introduction

Agent benchmarks evaluate systems that reason over multiple steps, invoke tools, and interact with environments. Reported behavior depends on the language model, scaffold, tools, prompts, execution limits, and task environment. Evaluation is also expensive: HAL reports 21,730 rollouts across nine benchmarks and roughly $40,000 in cost [Kapoor et al., 2026]. Cost-aware leaderboards therefore plot accuracy against dollar cost and recommend cost–accuracy frontiers.

A practical question follows: if a displayed model label is efficient on one agent benchmark, is it a promising low-cost choice elsewhere? Consider the observed GAIA–SWE-mini comparison: raw dollar-cost ranks are strongly aligned (Spearman $\rho = 0.89$ over 17 shared labels), yet aggregate dollars-per-expected-success ranks are only weakly aligned ($\rho = 0.29$); their resampled raw-minus-CPS contrast is 0.60 with a 95% sensitivity interval [0.14, 1.13]. A practitioner selecting a label solely because it is cheap on a GAIA-like leaderboard would therefore receive a useful raw-spend prior, but not a reliable ranking of expected dollars per successful SWE-like task. This is an aggregate ranking example, not a claim about any named model or production deployment. We study this question descriptively, holding the publicly displayed HAL Generalist Agent scaffold fixed and matching exact public model-label strings across workloads.

Our design is coverage-aware. Most HAL scaffolds occur on one benchmark, making a global model–scaffold decomposition unsupported. The Generalist Agent is the only broadly repeated scaffold. We retain six benchmarks with adequate pairwise overlap and analyze accuracy rank, dollar-cost rank, aggregate CostPerSuccess rank, and frontier membership only among shared labels. CostPerSuccess is defined as $C/\max(A, 0.01)$, where A is accuracy as a fraction; the supplement reports a 5% floor and zero-accuracy-exclusion sensitivities. It is an aggregate dollars-per-expected-success proxy, not observed successful-run cost. We contribute: (i) a cross-workload

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analysis separating robust high-overlap evidence from low-overlap estimates, (ii) configuration-resampling, Holm-corrected, and label-permutation checks, (iii) a cost-propensity adjustment, (iv) LOBO prediction of held-out raw cost ranks, and (v) frontier sensitivity across three definitions.

2 Related work

Kapoor et al. [2026] introduce HAL as a cost-aware leaderboard across coding, web, science, and customer-service tasks. Kapoor et al. [2024] argue for joint accuracy–cost evaluation of agents. SWE-Effi studies resource-constrained software-engineering agents and reports scaffold–model interaction within one domain [Fan et al., 2025]. Ndzomga [2026] study within-benchmark rank preservation under task reduction, while Benchmark^2 studies cross-benchmark rank consistency for static LLMs [Qian et al., 2026]. Open-SWE-Traces releases unencrypted SWE-agent and OpenHands trajectory corpora [Ahmad et al., 2026]. None directly establishes cross-workload rank transfer for interactive-agent dollar costs under a repeated scaffold.

3 Data and method

We analyze the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026]: 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 model labels. Accuracy and total dollar cost are complete, but 235 rows report one run. The frozen source, exact hash, and code are in the anonymous supplement.

The primary cohort fixes the displayed HAL Generalist Agent scaffold: 21 labels on CORE-Bench Hard, 17 on GAIA, 9 on SciCode, 7 on ScienceAgentBench, 18 on SWE-bench Verified Mini, and 14 on TAU-bench Airline. USACO has one Generalist result and is excluded. For each pair with at least five shared labels, we compute Spearman ρ and Kendall rank correlation for accuracy and dollar cost. We use 10,000 configuration-resampling draws for pairwise sensitivity intervals and 20,000 label-pairing permutations for residual-cost, CPS, and LOBO rank associations; these quantify finite-label sensitivity, not rollout-level or population-model uncertainty. Holm–Bonferroni correction is applied separately to 15 accuracy and 15 cost tests.

A nondominated point has no competitor that is at least as accurate and no more costly with one strict improvement. We also implement a HAL-inspired origin-anchored convex-hull reconstruction and a 5% tolerance sensitivity where dominance requires 5% lower cost and 5% higher accuracy. The tolerance is a near-tie stress test, not a calibrated measurement-error model.

4 Results

4.1 Raw cost-rank transfer is explained by stable label cost propensity

The robust raw correlations invite a confound check: cost rankings may encode persistent displayed-label price/cost tiers rather than portable execution dynamics. The frozen CSV lacks token components, provider alias, and historical price-card fields, so we cannot claim a list-price adjustment. We fit an observational pooled log-cost model on 86 Generalist rows with 30 parameters (25 label levels and six benchmark levels represented with reference coding), leaving 56 residual degrees of freedom. It explains 82.0 percent of log-cost variance, with adjusted R-squared of 72.7 percent. We treat this in-sample fit as descriptive rather than primary evidence; LOBO tests held-out rank prediction. The six high-overlap raw correlations collapse after residualization: none of the residual-cost rank correlations is significant under a 20,000-draw permutation test. Thus raw cost transfer largely reflects stable observed label-level cost propensity. This propensity may combine provider price tier, reasoning configuration, token use, cache treatment, routing, and unrecorded setup; it is not identifiable as list price from the public CSV. Exact adjusted coefficients are reported in the anonymous supplement. The raw cross-workload rank-transfer matrix, including each pair's Spearman correlation and shared-label count, is provided in the anonymous supplement. The raw dollar-cost structure is largely absorbed by stable displayed-label cost propensity.

4.2 Success adjustment reduces portability

Raw dollar cost does not answer which label buys expected success most cheaply. Under the primary one-percent denominator floor, CPS rank correlation survives 20,000-draw permutation testing and Holm correction for CORE–GAIA ($\rho = 0.87$, 95% sensitivity interval [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), and GAIA–TAU (0.78, [0.42, 0.94]); other pairwise CPS estimates have intervals crossing zero or fail Holm correction. The descriptive high-overlap mean is 0.59, versus 0.84 for raw cost, but we do not treat this average as a population-level contrast. The paired raw-minus-CPS resampling contrast is clearly positive only for GAIA–SWE-mini (difference 0.60, 95% interval [0.14, 1.13]); all pairwise inference and zero-exclusion/5% sensitivities are in the supplement.

Table 1: Higher-overlap pairs. Brackets are configuration-resampling sensitivity intervals; ND is nondominated-frontier Jaccard similarity on the shared-label cohort.

Pair	N	Accuracy ρ	Cost ρ	ND Jaccard
CORE-Bench Hard–GAIA	14	0.60 [0.16, 0.81]	0.93 [0.72, 0.99]	0.43
CORE-Bench Hard–SWE-bench Mini	14	0.48 [-0.05, 0.82]	0.74 [0.29, 0.93]	0.25
CORE-Bench Hard–TAU Airline	12	0.50 [-0.12, 0.86]	0.78 [0.26, 0.98]	0.17
GAIA–SWE-bench Mini	17	0.71 [0.31, 0.91]	0.89 [0.61, 0.99]	0.33
GAIA–TAU Airline	14	0.58 [0.14, 0.85]	0.86 [0.58, 0.97]	0.29
SWE-bench Mini–TAU Airline	14	0.76 [0.40, 0.91]	0.82 [0.46, 0.97]	0.43

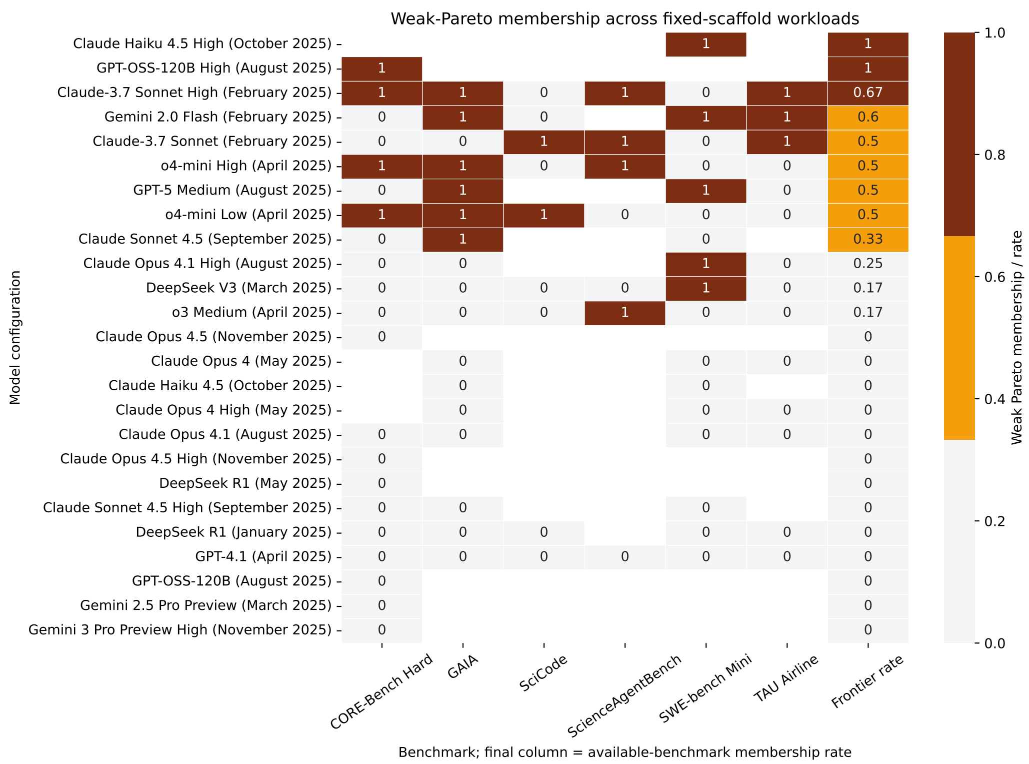

Figure 1: Nondominated cost–accuracy membership across primary workloads. The final column is each label's rate over available workloads. No broadly tested label appears on every available frontier.

4.3 Held-out portability is benchmark-specific

In leave-one-label-out checks, the positive LOBO findings are not driven by a single displayed label: CORE remains $\rho \in [0.73, 0.91]$, GAIA [0.84,0.92], SWE-mini [0.88,0.92], and TAU [0.79,0.95] across every omission. SciCode is unstable ([-0.50,0.14]); ScienceAgentBench remains directionally negative ([-0.77,-0.43]) but has only seven labels and a non-significant original permutation test. Observable aggregate diagnostics are descriptive only: SciCode has mean accuracy 2.1% with 33.3% zero-accuracy labels, while ScienceAgentBench has the narrowest log-cost dispersion. Six benchmarks cannot identify a domain-level cause.

4.4 Frontier membership is non-universal

No label tested on at least five workloads lies on every cost–accuracy frontier. This conclusion survives nondominance, the HAL-inspired hull, and 5% near-tie tolerance. The stronger statement that no label appears on more than 4/6 frontiers does not survive: under tolerance, o4-mini Low appears on 5/6. We therefore retain only the robust non-universal-winner conclusion.

4.5 Scaffold and frontier definitions matter

The source HAL labels recover 30 rows under the hull reconstruction, with two discrepancies; ordinary nondominance includes 19 additional points. A discrete buyer choosing one system and a deployer willing to randomize between systems therefore face different decision frontiers. Within individual benchmarks, scaffolds also differ substantially: across 15 shared labels on SWE-mini, displayed SWE-Agent averages 27.4 percentage points higher accuracy than the Generalist scaffold, with higher median cost. These contrasts motivate fixing the scaffold; they are not a general causal scaffold estimate.

5 Discussion and threats to validity

External matched trace extension. In 977 exact same-task/same-model pairs, the OpenHands/SWE-agent median ratio is 0.85 for trajectory turns (10,000-bootstrap interval [0.82,0.87]) and 0.85 for tool calls ([0.82,0.86]); OpenHands has lower values on 69.7% of pairs for each. Trajectory-character burden has a smaller but nonzero ratio effect, 0.95 ([0.92,0.98]), with OpenHands lower on 55.1% of pairs. Paired Wilcoxon tests are correspondingly small, while matched resolved status has no detected difference ($p = 0.197$). These are trace-burden proxies, not tokens or dollars, and the boundary sample is external to HAL.

A single benchmark can supply a useful prior about *raw* observed dollar cost for some held-out workloads, but the analysis does not establish portable workload-specific execution-cost dynamics: raw cost-rank transfer is largely absorbed by stable displayed-label cost propensity, and CostPerSuccess is less portable. Because this propensity may include provider price tier, reasoning configuration, token volume, cache treatment, routing, and unrecorded setup, it cannot be reduced to a list-price claim from the frozen public CSV. Practitioners should validate success-adjusted cost and token behavior on representative target tasks rather than treating one leaderboard frontier as a universal efficiency mechanism.

Construct validity is limited because public `Models` and `Agent Name` strings do not fully encode prompts, tools, budgets, harness versions, or pricing assumptions. The CSV also lacks provider aliases and token components, precluding historical list-price or common-tariff reconstruction; public HAL trace archives are encrypted. CostPerSuccess is an aggregate dollars-per-expected-success proxy, not task-level successful-run cost; the 1% floor is pre-specified and zero-exclusion/5% floor sensitivities are reported in the supplement. The external Open-SWE trace extension uses a reproducible four-shard boundary sample, not the full corpus, and its message characters/turns/tool calls are burden proxies rather than tokens or dollars. It is not merged with HAL or used to infer HAL mechanisms. The data cannot identify failure cost ratios, retries, token trajectories, or tail-dollar risk. Matching a display label under the Generalist scaffold is not a controlled base-model comparison. Internal validity is limited by sparse, unbalanced coverage and mostly single-run evaluations. DeepSeek R1 and Gemini 2.0 Flash are absent from ScienceAgentBench, but the public table provides no failure or exclusion reason; missingness is unidentifiable rather than assumed random. External validity is restricted to these snapshots and workloads.

6 Conclusion

Raw dollar-cost ranks transfer robustly across several high-overlap public HAL workload pairs, but pooled label and benchmark cost propensity explains most of this signal. Aggregate CostPerSuccess is less portable, showing that predictable cheapness is not equivalent to predictable dollars per expected success. An external exact task/model trace comparison further shows that scaffolds can materially change trajectory turns and tool calls without a detected success difference. The public evidence does not identify dollar-cost mechanisms, tokens, or provider prices, but it demonstrates why raw cost leaderboards combine stable label propensity with execution structure and should be validated with success-adjusted, trace-aware measures on target tasks.

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